1 The Six Trigonometric Functions

The six trigonometric functions are defined by coordinates on the unit circle and ratios of those coordinates. Their relationships determine their signs, domains, and graphs.

Let $P = (x, y)$ be the point at angle θ on the unit circle. Its coordinates are $x = \cos\theta$ and $y = \sin\theta$. The remaining four functions are formed from quotients and reciprocals of these coordinates.
Learning objectives. Define the six functions on the unit circle, calculate exact values in each quadrant, determine their domains, identify characteristic graph features, and apply trigonometric ratios to measurement problems.

1.1 From a point to six functions

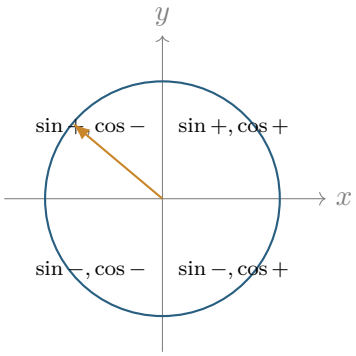
Last week we studied the unit-circle point $P = (x, y) = (\cos\theta, \sin\theta)$. The two coordinates determine four additional ratios whenever the denominators are nonzero:

$$\sin\theta = y, \quad \cos\theta = x, \quad \tan\theta = \frac{y}{x},$$
$$\csc\theta = \frac{1}{y}, \quad \sec\theta = \frac{1}{x}, \quad \cot\theta = \frac{x}{y}.$$

Thus the six functions encode related properties of a single point.

Definition 1.1. The *tangent* function is $\tan\theta = \sin\theta/\cos\theta$ where $\cos\theta \neq 0$. The functions csc, sec, cot are the reciprocals of sin, cos, tan respectively, wherever defined.

Example 1.1. At $\theta = \pi/3$ the circle point is $(1/2, \sqrt{3}/2)$. Thus $\tan\theta = \sqrt{3}$, $\sec\theta = 2$, $\csc\theta = 2/\sqrt{3} = 2\sqrt{3}/3$, and $\cot\theta = 1/\sqrt{3} = \sqrt{3}/3$.

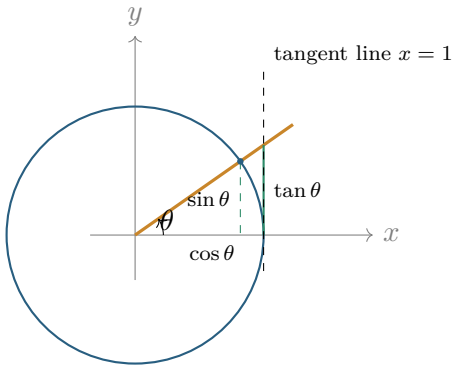


1.2 Tangent as a slope

For $\cos\theta \neq 0$, the line through the origin and $P = (\cos\theta, \sin\theta)$ has slope

$$m = \frac{\text{rise}}{\text{run}} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

Tangent therefore gives the slope of the terminal line. Its magnitude increases without bound as the line approaches a vertical position, at which the slope is undefined.



The terminal line intersects $x = 1$ at $(1, \tan\theta)$, whenever $\cos\theta \neq 0$. Hence tangent also has a geometric interpretation as a directed length on a line tangent to the unit circle.

Example 1.2. At $30°$, the unit-circle coordinates are $(\sqrt{3}/2, 1/2)$. The slope of the corresponding ray is

$$\tan 30° = \frac{1/2}{\sqrt{3}/2} = \frac{\sqrt{3}}{3}.$$

This is also the slope of a line rising one unit for every $\sqrt{3}$ units of horizontal displacement.

1.3 Signs and the quadrants

The coordinate signs determine every trig-function sign. In quadrant I both coordinates are positive; in II, only sine is positive; in III, tangent is positive because both coordinates are negative; and in IV, only cosine is positive. Reciprocal functions have the same signs as their corresponding original functions.

Example 1.3. For $\theta = 5\pi/4$, the reference angle is $\pi/4$ and $P = (-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore $\tan\theta = 1$, $\sec\theta = -\sqrt{2}$, and $\csc\theta = -\sqrt{2}$.

For a useful sign check, consider $150° = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$. Both sine and cosecant are positive, both cosine and secant are negative, and tangent and cotangent are negative:

$$\tan(5\pi/6) = -\frac{\sqrt{3}}{3}, \qquad \sec(5\pi/6) = -\frac{2\sqrt{3}}{3}, \qquad \csc(5\pi/6) = 2.$$

The signs of the six functions follow directly from the signs of sine and cosine.

	sin	cos	tan	csc	sec	cot
Quadrant I	+	+	+	+	+	+
Quadrant II	+	−	−	+	−	−
Quadrant III	−	−	+	−	−	+
Quadrant IV	−	+	−	−	+	−

Try it. The terminal point of an angle lies in quadrant IV and satisfies $\cos\theta = 3/5$. Determine $\sin\theta$, $\tan\theta$, and their reciprocals, showing how the quadrant tells you the signs.

1.4 Domains and undefined values

Division by zero is undefined. In particular, $\tan\theta$ and $\sec\theta$ do not exist where $\cos\theta = 0$, at $\pi/2 + k\pi$. Similarly $\cot\theta$ and $\csc\theta$ do not exist where $\sin\theta = 0$, at $k\pi$, for any integer k. Undefined values must be distinguished from zeros of a function.

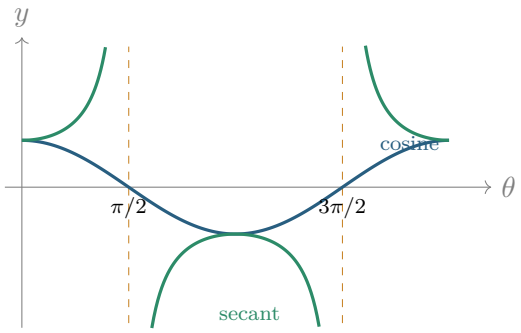
Try it. Find all six trig-function values at $\theta = 3\pi/2$, identifying every undefined value.

1.5 Zeros and vertical asymptotes

Near zeros of their denominators, trigonometric quotients may become arbitrarily large in magnitude. This behavior accounts for their vertical asymptotes.

For example, as θ approaches $\pi/2$ from the left, $\cos\theta$ approaches zero from above. Consequently $\sec\theta = 1/\cos\theta$ increases without bound. From the right, cosine is slightly negative, and secant becomes a large negative number. At $\pi/2$ itself, secant has no value.

The line $\theta = \pi/2$ is a vertical asymptote of both tangent and secant.



Only those portions of the secant graph within the displayed vertical range are shown. Its branches are separated by vertical asymptotes.

Try it. Describe the same phenomenon for $\csc\theta$ near $\theta = 0$. Explain why a zero of sine becomes an undefined value of cosecant.

1.6 Right-triangle ratios

For an acute angle θ in a right triangle, scale the triangle so its hypotenuse has length 1. The opposite and adjacent sides then give the coordinates $\sin\theta$ and $\cos\theta$. Restoring a general hypotenuse h gives

$$\sin\theta = \frac{\text{opposite}}{h}, \quad \cos\theta = \frac{\text{adjacent}}{h}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The unit-circle definitions extend the right-triangle ratios to arbitrary real angles.

Example 1.4. A ramp rises 1.5 meters over a horizontal run of 6 meters. Its angle of inclination satisfies $\tan\theta = 1.5/6 = 1/4$. Therefore $\theta = \arctan(1/4) \approx 14.0°$.

Writing o, a, and h for opposite side, adjacent side, and hypotenuse, respectively, gives

	opposite and hypotenuse	adjacent and hypotenuse	opposite and adjacent
direct	$\sin\theta = o/h$	$\cos\theta = a/h$	$\tan\theta = o/a$
reciprocal	$\csc\theta = h/o$	$\sec\theta = h/a$	$\cot\theta = a/o$

The six ratios describe the same triangle and agree with the unit-circle definitions for acute angles.

Example 1.5. A support cable stretches 13 meters from the ground to a point 12 meters above it. The horizontal run is $\sqrt{13^2 - 12^2} = 5$ meters. If θ is the angle with the ground, then

$$\sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = \frac{12}{5}, \quad \csc\theta = \frac{13}{12}, \quad \sec\theta = \frac{13}{5}, \quad \cot\theta = \frac{5}{12}.$$

Each quotient expresses a relation between lengths of the same triangle.

1.7 Exact values at special angles

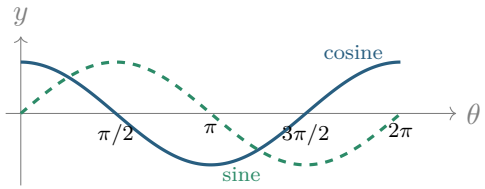
An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This produces $\sin 30° = 1/2$ and $\cos 30° = \sqrt{3}/2$. A square of side 1 has diagonal $\sqrt{2}$, producing $\sin 45° = \cos 45° = \sqrt{2}/2$.



1.8 Graphs of the trigonometric functions

On a trigonometric graph, the independent variable is an angle and the dependent variable is a coordinate or ratio at that angle. Such a graph is distinct from the circular trajectory of the associated point.

Sine and cosine remain between -1 and 1. Both repeat after a full rotation, so their fundamental period is 2π radians, or 360°. Tangent and cotangent repeat after half a turn, so their fundamental period is π radians, or 180°. Secant and cosecant share the 2π period of their reciprocal partners.



The zeros, excluded inputs, and fundamental period determine important features of a trigonometric graph.

Example 1.6. Rotation by π changes the signs of both unit-circle coordinates. Thus

$$\frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin\theta}{-\cos\theta} = \tan\theta.$$

Therefore tangent has period π on its domain.

The accompanying Six Trigonometric Graphs and Reciprocal Trigonometric Graphs demonstrations illustrate these functions and their asymptotes.

Try it. Sketch $y = \tan x$ and $y = \cot x$ on $(0, \pi)$, marking their zeros and asymptotes. How do the two plots differ at the interval's endpoints?

1.9 Applications: orientation and measurement

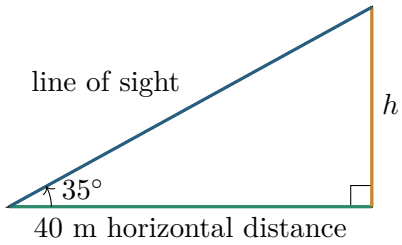
Suppose an observer stands 40 m from the foot of a vertical tower. The angle of elevation to its top is 35°. The tower's height h satisfies

$$\tan 35^\circ = \frac{h}{40}, \qquad h = 40 \tan 35^\circ \approx 28.0 \text{ m}.$$

Here the tangent function links an angle to a ratio of lengths. Always sketch the physical situation and label which distances are opposite and adjacent before calculating.

1.10 Measurement with trigonometric ratios

The appropriate trigonometric ratio depends on the available lengths. Tangent relates height to horizontal distance; sine relates height to line-of-sight distance.



Worked example: height measurement. An observer is 20 meters from the base of a tree and measures an angle of elevation of 40° to its top. If the observer's eyes are 1.6 meters above ground, the elevation of the treetop above eye level is

$$h_{\text{above eyes}} = 20 \tan 40^\circ \approx 16.8 \text{ m}.$$

The tree height is therefore about $16.8 + 1.6 = 18.4$ meters. The observer's eye height must be added to determine the height above ground.

Try it. A rescue drone is observed at an elevation angle of 25° from an observer whose eye level is 1.7 meters above ground. The point directly beneath the drone is 90 meters away horizontally. Estimate the drone's altitude.

1.11 Exercises

1. At $\theta = 2\pi/3$, find all six functions exactly.
2. At $\theta = 7\pi/4$, find $\tan, \cot, \sec, \csc$.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.
4. For which angles in $[0, 2\pi)$ is $\csc\theta$ undefined?
5. A right triangle has opposite side 5 and adjacent side 12. Find all six trig ratios exactly.
6. A ladder leans against a wall, reaching 8 m high. Its foot is 3 m from the wall. Find the angle with the ground.
7. A drone is viewed at an elevation angle of 28° from a point 75 m horizontally from the point below it. Estimate its altitude.

8. Show that $\sin^2\theta + \cos^2\theta = 1$ implies $1 + \tan^2\theta = \sec^2\theta$ wherever $\cos\theta \neq 0$.
9. Find the exact values of all six trig functions at $11\pi/6$.
10. Describe the zeros and vertical asymptotes of $y = \tan x$ on $[-2\pi, 2\pi]$.
11. Sketch one period of $y = \sec x$, labeling asymptotes and its lowest positive and highest negative values.
12. Explain why neither $\csc x$ nor $\sec x$ can equal $1/2$.
13. If $\cos\theta = -12/13$ and θ lies in quadrant II, determine all other trig values exactly.
14. Show using the unit circle that $\tan(\theta + \pi) = \tan\theta$ wherever both sides are defined.
15. A point on a rotating wheel has unit-circle coordinates $(0, -1)$. Identify the angles in $[0, 2\pi)$ and all four undefined or zero-valued functions.